WORKED SOLUTION

STEP II 2018, Question 6

Pure • Factorials and Primes

STEP II 2018, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s2-q6

The key idea

Both parts turn on one fact: a prime p divides $k!$ exactly when $p \leq k$. In (i), once $n \geq 5$ the factorial is a multiple of 5, and so is $n! + 5$. In (ii), the left-hand side is built only from numbers up to $2n-1$, so no prime bigger than $2n-1$ can divide it. If $m!$ is large enough to contain such a prime, the equation fails. The two theorems are there to make m large and to supply that prime.

Solution

Part (i)

For $n \geq 5$, 5 is one of the factors of $n!$, so $5 \mid n!$ and hence $5 \mid n! + 5$. But $n! + 5 > 5$, so $n! + 5$ has a factor 5 strictly between 1 and itself, and is not prime. So only $n \leq 4$ needs checking:

$$1! + 5 = 6, \qquad 2! + 5 = 7, \qquad 3! + 5 = 11, \qquad 4! + 5 = 29.$$

The number 6 is not prime, and the other three are. The solutions are

$$(n, p) = (2, 7),\ (3, 11),\ (4, 29).$$

Part (ii)

Write $P_n = 1! \times 3! \times \cdots \times (2n-1)!$. Every factor in every one of these factorials is an integer between 1 and $2n-1$. So if a prime q divides P_n then, since q is prime, it divides one of those integers, and so $q \leq 2n-1$. That gives the key fact:

$$\text{no prime } q > 2n-1 \text{ divides } P_n. \qquad (*)$$

On the other hand, every prime $q \leq m$ divides $m!$. So if $P_n = m!$, then m is smaller than every prime greater than $2n-1$.

Large n. Suppose $n \geq 7$ and $P_n = m!$. By Theorem 1, $m! = P_n > (4n)!$, so $m > 4n$ (factorials increase). By Theorem 2 there is a prime q with $2n < q < 4n$. Then $q < 4n < m$, so $q \mid m! = P_n$. But $q > 2n > 2n - 1$, so this contradicts $(*)$. So there are no solutions with $n \geq 7$.

Ask yourself Why can't we just quote the theorems for every n?

Theorem 1 is only claimed for $n \geq 7$, and in fact it is false for $n = 6$. So $n = 1, \ldots, 6$ need to be checked by hand. We use $(*)$ again to keep the arithmetic short.

Small n.

- $n = 1$: $P_1 = 1 = 1!$, so $m = 1$.
- $n = 2$: $P_2 = 1 \times 6 = 6 = 3!$, so $m = 3$.
- $n = 3$: $P_3 = 6 \times 120 = 720 = 6!$, so $m = 6$.
- $n = 4$: $P_4 = 720 \times 5040$. Since $720 = 6!$ and $5040 = 7 \times 8 \times 9 \times 10$, we get $P_4 = 6! \times 7 \times 8 \times 9 \times 10 = 10!$, so $m = 10$.
- $n = 5$: $P_5 = P_4 \times 9! = 10! \times 9! > 10!$, so $m \geq 11$. Then the prime 11 divides $m!$, but $11 > 9 = 2n - 1$, contradicting $(*)$. No solution.
- $n = 6$: $P_6 = 10! \times 9! \times 11! > 12!$ (since $10! \times 9! > 12$), so $m \geq 13$. Then the prime 13 divides $m!$, but $13 > 11 = 2n - 1$, contradicting $(*)$. No solution.

In each successful case m is unique, because factorials of positive integers are strictly increasing from 2! onwards (and $1! = 1 < 2!$). So the solutions are

$$(n, m) = (1, 1),\ (2, 3),\ (3, 6),\ (4, 10).$$

Checks

The identity $1!\,3!\,5!\,7! = 10!$ is easy to confirm: both sides equal 3 628 800. A computer search over $n \leq 59$ and $m < 400$ found exactly the four pairs above, and confirmed that $n! + 5$ is prime only for $n = 2, 3, 4$ (searching $n \leq 30$). It also confirms that Theorem 1 fails at $n = 6$, which is why the hand check of $n = 5$ and $n = 6$ really is needed.

Common slips

- Forgetting $n = 1$ in part (ii): the product then has the single factor 1!, so $(1, 1)$ is a solution.
- Using Theorem 1 for $n = 5$ or $n = 6$. It is stated only for $n \geq 7$, so those cases need their own argument.
- Saying "$q \mid m!$" without checking $q \leq m$. That is exactly why we need $m > 4n$ from Theorem 1.

Beyond STEP

Theorem 2 is Bertrand's postulate: for every $n \geq 1$ there is a prime p with $n < p \leq 2n$. Chebyshev proved it in 1852, and Erdős later gave a beautiful elementary proof using the central binomial coefficient $\binom{2n}{n}$. Equations such as $n! + 1 = m^2$ (Brocard's problem) are still open. Davenport's *The Higher Arithmetic* is a good next step into the arithmetic of primes and factorials.

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